Swayam Shah

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EDUCATION

University of California, Los Angeles (UCLA) — Expected June 2028
Bachelors of Science, Major: Mechanical Engineering — *Los Angeles, CA*

- GPA: 3.922 / 4.000
- Relevant Coursework: Statics, Dynamics, Strength of Materials, Thermodynamics, Fluid Mechanics, Electrical Circuits

SKILLS

Mechanical Design & Analysis: SolidWorks, GD&T, Tolerance Analysis, DFM/DFA, ANSYS (FEA: stress, buckling, fatigue)
Manufacturing: CNC & Manual Mill/Lathe, 3D Printing, Plasma/Laser Cutting, Welding, Soldering, Assembly
Electrical/Software: MATLAB, C++, Python, Java

EXPERIENCE

Bruin Baja Racing (SAE) — September 2025 - Present
Destructive Testing Engineer — *Los Angeles, CA*

- Developed a gearbox fatigue test system to characterize drivetrain durability under race-representative loading and reduce uncertainty in fatigue-life predictions.
- Designed, fabricated, and assembled a drivetrain test stand integrating engine, gearbox, and brake subsystems to reproduce cyclic loading and braking-induced torque transients.
- Integrated Hall-effect and brake-line pressure instrumentation to track rotational speed, estimate applied torque, and enable reliable load monitoring over 100k+ cycles.
- Correlated experimental results with FEA and Miner's Rule fatigue modeling to identify tooth root bending stress as the dominant fatigue failure and guide future gearbox durability and mass optimization decisions.

Bruin Formula Racing (FSAE) — September 2025 - Present
Suspension & Vehicle Dynamics Subteam Member — *Los Angeles, CA*

- Redesigned front suspension A-arms by replacing 4130 steel members with a carbon-fiber composite, improving structural efficiency while achieving 20% mass reduction.
- Performed structural analyses across braking and combined cornering load cases to evaluate stress distributions, buckling resistance, and design margins under competition loading.
- Optimized composite tube geometries through OD/ID trade studies and material selection, choosing uni-rolled carbon fiber based on strength-to-weight and cost-performance considerations.
- Designed 6061-T6 aluminum plugs and wafer interfaces in SolidWorks to transfer suspension loads into composite members while satisfying manufacturability and assembly requirements.

UCLA Bionics Research Lab — September 2025 - Present
Undergraduate Research Intern — *Los Angeles, CA*

- Designed and modeled an actuated colonoscopy snare in SolidWorks, developing and refining the mechanism through multiple design iterations.
- Evaluated different materials to replicate the flexibility and mechanical behavior of a colonoscope, allowing realistic testing of the device.
- Built and tested a large-scale prototype to verify functionality, identify design issues, and guide development of a micro-scale version for medical applications.

PROJECTS

Real World Design Challenge - Team CALtronics

- Designed and modeled wings, fuselage, and landing gear for a 4,000 lb unmanned cargo aircraft, performing structural analyses and system-level trade studies to support a 355-mile rural cargo mission.
- Evaluated aircraft stability, weight distribution, and structural loading while balancing performance, manufacturability, and mission requirements throughout the design process.
- Collaborated with FAA engineers during technical design reviews, refining the aircraft configuration and contributing to a 1st-place finish in California and 3rd place out of 455 teams nationally.